

PRACTICAL - WEEK 1 AND 2

Program 1:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    char cmd[100];
    printf("Enter linux command:");
    scanf("%s", cmd);
    pid_t pid = fork();
    if (pid==0){
        printf("Child pid: %d\n",getpid());
        execlp(cmd, cmd, NULL);
        perror("Exec Failed");
    }
    else if (pid > 0){
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
userhasini@DESKTOP-QDSCC4:~$ nano child.c
userhasini@DESKTOP-QDSCC4:~$ gcc child.c -o child
userhasini@DESKTOP-QDSCC4:~$ ./child
Enter linux command:ls
Parent PID: 649
child pid: 650
child  display  file1.txt  hello.c  parent  practical1.c
child.c  display.c  hello    oslab    parent.c  practical1.c.save
Child process completed.
userhasini@DESKTOP-QDSCC4:~$
```

Program 2:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main(){
    int src, dest;
    char buffer[1024];
    int bytes;
    src = open("input.txt", O_RDONLY);
    if(src < 0){
        printf("Cannot open source file\n");
        return 1;
    }
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }
    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

```
userhasini@DESKTOP-QDSCC4:~$ nano rdonly.c
userhasini@DESKTOP-QDSCC4:~$ gcc rdonly.c -o rdonly
userhasini@DESKTOP-QDSCC4:~$ ./rdonly
-bash: ./: Is a directory
```